Attack-Method Choice Changes Robustness Conclusions for a Standard LSTM Rainfall-Runoff Model

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Key Points:

- Iterative adversarial perturbations cause 16 times the median NSE loss of matched random noise for a 531-basin CAMELS-US LSTM at ε=0.1.
- Vulnerability is universal but scales with hydrological memory: the snow-dominated catchments the model fits better are damaged most.
- Perturbations preserving each feature's mean and standard deviation evade moment-based quality control yet retain 78% of the damage.

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Abstract
Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong predictive skill in rainfall-runoff modeling. As these models move toward operational deployment, their robustness to input uncertainty becomes a critical concern. Yet existing assessments often rely on random noise or single-step adversarial methods (FGSM), which may underestimate worst-case vulnerability. Here we present a multi-level adversarial stress test of a 531-basin CAMELS-US benchmark LSTM, applying a spectrum of perturbation methods from random noise through iterative optimization, three constraint levels of increasing plausibility, targeted attacks on flood and pre-event windows, and a minimum-perturbation analysis. These perturbations are not intended to reproduce the empirical distribution of real-world forcing errors; rather, they serve as constrained worst-case stress tests within observation-error-scale uncertainty budgets. We find that the model is substantially more fragile than random-noise testing suggests: at $\varepsilon = 0.1$, iterative scheduled-step PGD causes a median NSE degradation of 0.38, against 0.024 for a structure-matched random-sign search of equal sampling budget, a ratio of about 16:1. Single-step FGSM gives a similar but weaker median degradation of 0.30, so the dominant conclusion change is between random and gradient-based testing rather than between single-step and iterative gradient methods. Perturbations that preserve the per-feature mean and standard deviation of the inputs evade moment-based quality control by construction yet still retain 78% of the unconstrained degradation. The vulnerability is universal but uneven: every catchment type degrades—even the most robust, rain-driven catchments lose a median of 0.27 in NSE—yet its severity scales with hydrological memory. Snow fraction is the strongest catchment predictor of adversarial damage (rank correlation 0.61 after controlling for clean-data skill), and the snow-dominated catchments the model fits better are damaged most, so the stress test doubles as a probe of which learned hydrological dependencies are least robust. In snow catchments the attack gains leverage on the temperature (snowmelt) forcing and suppresses predicted flood peaks about twice as strongly as in rain catchments, consistent with a brittle reliance on learned long-memory integration. Robustness assessment for standard hydrological deep-learning configurations therefore requires multi-level stress testing: random-noise sensitivity analysis substantially underestimates the vulnerability that gradient-based adversarial attacks reveal.

Plain Language Summary

LSTM neural networks are increasingly used for streamflow prediction and are moving toward operational use. We tested how sensitive a standard 531-basin CAMELS-US benchmark LSTM rainfall-runoff model is to small but structured errors in its weather inputs, within realistic uncertainty bounds. Using an optimization method that follows the model's own gradient, we searched for the most harmful input errors allowed by those bounds. These perturbations are not meant to mimic the most common real-world errors; they are stress tests that ask how badly the model can fail if the input errors happen to align with its sensitivities. Across the 531 basins, these constrained worst-case perturbations were about sixteen times more damaging than random weather noise of the same size and search effort. A simple single-step gradient method found much of the same vulnerability, so the key difference is between using the model's gradient at all and not using it. We also found that some harmful perturbations would slip past routine data-quality checks that only compare the mean and spread of each variable to climatology. These results do not argue against deploying LSTM models, but they show that standard hydrological deep-learning configurations need stronger stress testing than random-noise sensitivity analysis alone.

1 Introduction

Long Short-Term Memory (LSTM) networks have transformed hydrological prediction over the past decade. Beginning with the pioneering work of Kratzert et al. (2018), deep learning models have consistently matched or exceeded the performance of process-based hydrological models on standardized benchmarks such as CAMELS (Addor et al., 2017), achieving median Nash-Sutcliffe Efficiency (NSE) values above 0.7 across hundreds of catchments (Kratzert et al., 2019; Gauch et al., 2021). These advances have catalyzed growing interest in deploying LSTM-based models in operational forecasting systems (Nearing et al., 2021).

As deployment moves forward, a fundamental question demands attention: how reliable are these models when confronted with imperfect input data? In practice, meteorological forcings are never exactly known. Rain gauge measurements carry systematic biases of 5–15% (Sevruk, 1982; Groisman & Legates, 1994). Gridded products such as Daymet, Maurer, ERA5, and CHIRPS introduce additional uncertainty through spatial interpolation, with temperature errors of 1–2 °C and precipitation errors that can exceed 2 mm/day in complex terrain (Thornton et al., 2021; Hersbach et al., 2020). If model predictions are highly sensitive to such perturbations, then predictive accuracy on clean benchmarks may overstate operational reliability.

The conventional approach to testing input sensitivity, adding random Gaussian noise to the forcings, provides a measure of average-case robustness. However, random perturbations distribute their probe equally in all directions and are unlikely to land on the particular forcing-error pattern to which the trained model is most sensitive. Adversarial robustness analysis, borrowed from the machine-learning security literature (Goodfellow et al., 2015; Madry et al., 2018), offers a more systematic alternative: it uses gradient information from the trained model itself to identify, within a given uncertainty budget, the forcing perturbation that would most damage the streamflow prediction, and so provides a lower bound on worst-case performance. In this study, such perturbations are not treated as literal samples from the empirical forcing-error distribution. They are constrained stress tests that ask whether harmful error patterns exist at all within plausible uncertainty budgets, not whether such patterns dominate observed forcing errors.

Yang et al. (2026) applied adversarial robustness analysis to hydrological models using the Fast Gradient Sign Method (FGSM; Goodfellow et al., 2015), evaluating LSTM and HBV models across 1,347 German catchments from the CAMELS-DE dataset. Their findings were encouraging: LSTMs demonstrated greater robustness than HBV models, catastrophic failure was reported as rare, and model responses scaled approximately linearly with perturbation magnitude.

However, FGSM is a single-step attack: it evaluates the model's gradient once at the clean forcing and takes one maximum-sized push along the resulting sign direction. While computationally efficient, single-step methods are known in the adversarial machine-learning literature to systematically overestimate model robustness, because they implicitly linearize the model's response to forcing perturbations around the clean operating point. Iterative methods such as Projected Gradient Descent (PGD; Madry et al., 2018) and its scheduled-step variants (Croce & Hein, 2020) refine the perturbation over many smaller steps and so capture the nonlinearities in the model's response, generally finding more harmful forcing patterns than single-step probes. The gap between single-step and iterative methods is not merely quantitative; it can qualitatively change conclusions about whether a model is "robust" or "vulnerable" (Carlini et al., 2019). Yang et al. (2026) themselves noted the need for additional test scenarios and perturbation methods. This raises a question we take up here: how fragile is a standard LSTM rainfall-runoff configuration when subjected to systematic, multi-level adversarial stress testing, and which methodological choices most change the conclusions drawn?

To answer this question, we construct a multi-level adversarial stress-testing framework and apply it to a standard LSTM model on the 531-basin CAMELS-US benchmark. The framework comprises three components: (i) a *perturbation spectrum* spanning methods of increasing optimization strength, from random noise through single-step FGSM to iterative scheduled-step PGD; (ii) a *constraint hierarchy* controlling perturbation plausibility, including a statistical constraint that preserves input-feature means and standard deviations; and (iii) *operationally motivated analyses*, comprising targeted flood/low-flow attacks, a causal-trigger analysis restricting perturbations to pre-event windows, a minimum-perturbation analysis quantifying per-catchment safety margins, and a distributional-detectability test. While Yang et al. (2026) focused on comparing model architectures under a single attack method, our study takes a complementary perspective: we hold the model fixed and systematically vary the attack to characterize how the strength and structure of the stress test affect the conclusions drawn about model robustness. We further ask what the resulting vulnerability reveals about the model itself: by relating per-catchment adversarial damage to hydrological attributes, we test whether the fragility is uniform or concentrated in particular hydroclimatic regimes, and find that it scales with hydrological memory—largest in snow-dominated, storage-driven catchments and smallest in flashy, rainfall-driven ones—so the adversarial stress test doubles as a diagnostic of which learned input dependencies are least robust.

2 Methods

2.1 Study Area and Target Model

We evaluate a single-layer LSTM model (Kratzert et al., 2018) with 256 hidden units and 27 static catchment attributes, implemented in the neuralhydrology framework (Kratzert et al., 2022). We deliberately use this standard configuration as the victim model so that the analysis isolates attack-method effects rather than architectural differences. The model was trained on the CAMELS-US dataset (Newman et al., 2015; Addor et al., 2017) with five dynamic input features from the Maurer forcing product: precipitation (mm/day), minimum and maximum temperature (°C), shortwave radiation (W/m^2), and vapor pressure (Pa). In the Maurer product the daily minimum and maximum temperature inputs are identical, so the five raw channels carry four independent dynamic degrees of freedom; the perturbation treats the two temperature channels as a single shared variable throughout (Section 2.2).

The target model reproduces the 531-basin CAMELS-US benchmark of Kratzert et al. (2019): the same 531 basins are used for training and testing over disjoint periods. We use the dataset's reverse temporal split, training on 1 October 1999–30 September 2008 and testing on 1 October 1989–30 September 1999, with a 270-day input sequence and a single-step (last-day) prediction target. The trained model attains a test-period median NSE of 0.733 across the 531 basins, matching the benchmark of Kratzert et al. (2019). Adversarial evaluation pins the epoch-30 checkpoint; reproducibility details and run paths are provided in Supporting Information Text S1.

2.2 Perturbation Methods

We employ perturbation methods that span a spectrum of increasing optimization strength (Table 1). All methods operate on the standardized forcing time series, with each variable rescaled to zero mean and unit variance over the training period. The perturbation budget ε caps the shift, in standardized units, of any single timestep–variable component. Table 2 translates ε back into physical units (mm/day, °C, etc.). The attack perturbs a single shared forcing series per basin: the same per-day, per-feature perturbation is applied to every overlapping input window, so the result is a realizable tampering of the forcing record rather than an independent per-window perturbation. Because the two Maurer temperature channels are identical, they are perturbed together as one variable in every method.

This design is diagnostic rather than predictive. The random and adversarial perturbations are not intended to reproduce the empirical distribution of operational forcing errors; they test whether, within a specified uncertainty budget and plausibility constraints, structured input errors that materially degrade model performance can be constructed at all.

Table 1 — Perturbation methods used in this study, ordered by optimization strength.

Method	Type	Gradient	Iter.	Role
Random sign	Random baseline	None	0	Structure-matched control
Gaussian noise	Random baseline	None	0	Interior random noise
FGSM	Single-step adv.	Single	1	Bridge; used by Yang et al. (2026)
Scheduled-step PGD	Iterative adv.	Iterative	50	Worst-case lower bound
Min. perturbation	Safety margin	Iterative	–	Per-basin L_2 margin
Causal trigger	Temporally constr.	Iterative	50	Pre-event vulnerability

2.2.1 Random Noise Baselines

Two random perturbation methods serve as baselines. For each we draw 100 independent random samples per evaluation and retain the sample with the largest NSE drop, so that each baseline reflects the worst case identified by a light random search rather than a single arbitrary draw. *Random sign* sets each timestep–variable component to $\pm\varepsilon$ with equal probability; because the worst-case linear perturbation under an L_∞ budget lies at such a $\pm\varepsilon$ corner, random sign occupies the same $\pm\varepsilon$ corners of the budget as the gradient attacks and is therefore the structure-matched random control. *Gaussian noise* samples i.i.d. values from $\mathcal{N}(0, \varepsilon^2)$ at every timestep and forcing variable, clipped to $[-\varepsilon, \varepsilon]$, and represents white sensor noise that places most of its energy in the interior of the budget. Neither uses gradient information from the model. We use random sign as the primary random baseline because it isolates the effect of using the gradient from the unrelated effect of where in the budget the perturbation places its energy; Gaussian noise is reported alongside it to show that the interior-versus-corner placement changes the random damage only marginally relative to the gradient gap.

2.2.2 FGSM (Single-Step Adversarial)

The Fast Gradient Sign Method (FGSM) seeks the worst single-step perturbation within the budget. Backpropagation through the trained model returns, at every timestep and for every forcing variable, the partial derivative of the loss with respect to that input value. The sign of this derivative indicates whether shifting the value upward or downward degrades the prediction; FGSM applies a perturbation of magnitude ε in this unfavorable direction at every component simultaneously,

$$\mathbf{x}_{\text{adv}} = \mathbf{x} + \varepsilon \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}), \mathbf{y})) \,, \tag{1}$$

where $\mathcal{L}$ is the negative NSE loss, f is the trained model, and $\mathbf{y}$ is the observed streamflow. FGSM requires a single forward pass and a single backward pass, and corresponds to the worst forcing-error pattern that a linearization of the model around the clean input would predict.

2.2.3 Scheduled-Step PGD (Iterative Adversarial)

FGSM implicitly linearizes the model around the clean forcing. For small budgets this linearization is accurate, and a single sign step nearly saturates the achievable damage. For larger budgets the model's response curves away from the linear approximation within the ε neighborhood, so the direction picked at the clean input is no longer the worst direction within the budget. We therefore use an iterative scheduled-step projected gradient method (Croce & Hein, 2020): rather than one step of size ε, it takes many smaller steps, recomputes the gradient at each new candidate forcing, and projects the result back into the budget whenever a component would otherwise leave it,

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{B}(\mathbf{x},\varepsilon)}\left[\mathbf{x}_t + \alpha_t \cdot \mathrm{sign}(\nabla_{\mathbf{x}}\mathcal{L}(f(\mathbf{x}_t), \mathbf{y}))\right]. \tag{2}$$

Here $\Pi_{\mathcal{B}(\mathbf{x},\varepsilon)}$ denotes projection onto the ε-neighborhood around the clean input, implemented by clipping any component that exceeds the budget back to the boundary, and α_t is a step size that starts at 2ε and is halved at preset checkpoints. We use 50 iterations from a single random initialization within the ε-neighborhood. This is a deliberately conservative iterative attack: it uses no momentum and no multi-restart, so it is, if anything, a lower bound on what a stronger iterative method could achieve. We refer to it as scheduled-step PGD throughout.

2.2.4 Minimum-Perturbation (Safety Margin) Attack

Whereas FGSM and scheduled-step PGD quantify the maximum damage achievable under a fixed perturbation budget, the minimum-perturbation analysis quantifies the dual quantity: the smallest perturbation, measured by its L_2 norm in standardized units, required to drive a basin's test-period NSE below zero. This converts the analysis from an attack-magnitude axis to a per-catchment safety-margin axis. We compute it by bisection on the L_2 radius: for each candidate radius we maximize damage within the L_2 ball by projected gradient ascent and test whether NSE crosses zero, then bisect to the smallest breaking radius. Catchments that require a large L_2 perturbation before NSE drops below zero have a wide margin against forcing error of any structure; catchments broken by a small L_2 perturbation are operationally fragile. This analysis is run on a 76-basin subset (every seventh benchmark basin) for computational cost.

2.2.5 Causal Trigger Attack

The causal trigger attack quantifies how much of the iterative damage is achievable by perturbations confined to the period most relevant for flood early warning. The days immediately preceding a flood peak are operationally consequential: forecast skill is most valuable in this window, and input data are most often degraded by sensor failure or telemetry latency. We identify flood peaks as local maxima of the observed discharge exceeding its 95th percentile, separated by at least 14 days (a median of 46 peaks per basin). Perturbations are restricted to a window of 1, 3, 7, or 14 days before each identified peak; outside these windows the forcing is unperturbed. Within the perturbation windows, scheduled-step PGD optimizes the squared error of the peak-day predictions. We report two quantities: the degradation of an NSE computed on the peak days alone ($\Delta\mathrm{NSE}_{\mathrm{peak}}$), which measures how badly the flood-peak predictions themselves can be corrupted from pre-event days, and the degradation of the overall test-period NSE ($\Delta\mathrm{NSE}_{\mathrm{overall}}$), which measures the share of total vulnerability reachable from pre-event windows alone. At a 14-day window the pre-event interval can reach a preceding peak, so the windows are not strictly disjoint at the longest setting.

2.3 Constraint Hierarchy

The same ε budget can describe forcing-error patterns of very different physical realism. To make this explicit, we evaluate scheduled-step PGD under three nested constraints, ordered from most permissive to most restrictive.

The first level, the *magnitude constraint*, only requires that no single timestep–variable component be perturbed by more than ε in standardized units. This is the default constraint used throughout the perturbation literature; it admits forcing patterns that may be physically implausible (negative precipitation, unrealistically extreme temperatures) so long as each individual deviation stays within the budget.

The second level, the *physical constraint*, additionally clips each forcing variable to a physically admissible range: non-negative precipitation, shortwave radiation bounded below the solar constant, non-negative vapor pressure, and temperatures bounded to climatologically reasonable extremes, following Yang et al. (2026). This rules out patently unphysical forcing patterns while leaving the magnitude budget unchanged.

The third level, the *statistical constraint*, additionally requires the perturbed forcing series to preserve, for each variable, the same mean and standard deviation as the clean series over the whole evaluation period. This is intended as a proxy for input data that has already passed routine quality-control checks comparing summary statistics to climatology: any QC pipeline based on per-variable means and standard deviations would, by construction, fail to flag perturbations satisfying this constraint. The constraint pins the whole-period marginal moments; it does not pin the moments of individual 270-day input windows, which the recurrent model actually consumes, nor higher moments or temporal ordering.

2.4 Mapping ε to Physical Units

Table 2 maps ε values to physical units using training-period standard deviations. At $\varepsilon = 0.1$, perturbations correspond to approximately ± 0.7 mm/day for precipitation and ± 1.1 °C for temperature, values within the typical error range of gridded products such as Maurer (Maurer et al., 2002; Thornton et al., 2021). Our primary analyses focus on $\varepsilon = 0.1$ as a realistic worst-case scenario.

Table 2 — Mapping between ε and physical units for each input feature. Training-period standard deviations (σ, Maurer forcing) are used for conversion. The two temperature channels are identical in the Maurer product and share a single σ.

Feature	σ	ε=0.05	ε=0.1	ε=0.2	Typical obs. error
Precip. (mm/day)	6.96	±0.35	±0.70	±1.39	Gauge: 0.2–0.5
$T_{\max}$=$T_{\min}$ (°C)	10.73	±0.54	±1.07	±2.15	Station: 0.2–0.5
Radiation (W/m^2)	131.18	±6.6	±13.1	±26.2	Pyranometer: 5–10
Vap. pres. (Pa)	631.72	±31.6	±63.2	±126	Hygrometer: 20–50

2.5 Evaluation Metrics

Our primary robustness metric is $\Delta\text{NSE} = \text{NSE}_{\text{adv}} - \text{NSE}_{\text{clean}}$, where negative values indicate degradation caused by the perturbation. Because the per-basin distribution of ΔNSE is heavy-tailed toward harm, we report medians and interquartile ranges across catchments rather than means; we treat the median ΔNSE as the headline statistic, because the fraction of basins driven below a fixed NSE threshold conflates attack damage with baseline skill. Supplementary metrics include ΔKGE for the Kling–Gupta Efficiency and a Kolmogorov–Smirnov (KS) two-sample p-value comparing the clean and perturbed marginal distribution of each forcing variable, used as a measure of how detectable a perturbation would be under a routine distributional QC check. The NSE computed by the attack on the concatenated last-day predictions matches the official benchmark per-basin NSE to within $3{\times}10^{-7}$. All 531 benchmark basins yield finite NSE and ΔNSE over the test period, so all headline statistics are reported over the full 531 basins; the constraint-ablation, targeted, causal, and detectability analyses use a 107-basin subset (every fifth benchmark basin) for cost.

3 Results

3.1 Attack-Method Comparison and Vulnerability Distribution

Table 3 reports median ΔNSE across the 531 basins for each attack method at four perturbation budgets. Three features merit comment: (i) the dominant gap is between methods that use the trained model's gradient (FGSM, scheduled-step PGD) and methods that perturb the forcing without reference to the model (the random baselines), rather than between iterative and single-step gradient methods; (ii) the gap between single-step FGSM and iterative PGD increases monotonically with ε; and (iii) the underlying ΔNSE distribution is heavy-tailed, with the at-risk tail widening with ε.

Table 3 — Median ΔNSE [Q25, Q75] across the 531 benchmark basins for each attack method (magnitude constraint, untargeted). The last row gives the ratio of median scheduled-step-PGD degradation to median random-sign degradation, the structure-matched random control.

Method	N	ε=0.01	ε=0.05	ε=0.1	ε=0.2
Scheduled-step PGD	531	−0.026 [−0.034, −0.018]	−0.153 [−0.209, −0.106]	−0.381 [−0.539, −0.251]	−1.097 [−1.714,
FGSM	531	−0.025 [−0.034, −0.018]	−0.141 [−0.196, −0.099]	−0.301 [−0.448, −0.197]	−0.643 [−1.107,
Random sign	531	−0.0019	−0.0107	−0.0239	−0.0592
Gaussian	531	−0.0014	−0.0075	−0.0162	−0.0378
PGD / random sign		13.2×	14.2×	15.9×	18.5×

At $\varepsilon = 0.1$, the median ΔNSE under scheduled-step PGD is −0.381, against −0.301 for FGSM and −0.024 for random sign. Both random baselines are an order of magnitude weaker than either gradient method (random sign −0.024, Gaussian −0.016), and they differ from each other by far less than either differs from the gradient attacks, confirming that the dominant factor distinguishing methods is the use of gradient information rather than where in the budget a model-blind perturbation places its energy. The PGD-to-random-sign ratio at $\varepsilon = 0.1$ is 15.9×, more than an order of magnitude larger than the PGD-to-FGSM ratio of 1.27×. The ratio of medians and the median of per-basin ratios agree to within one unit, and scheduled-step PGD exceeds random sign for every one of the 531 basins at every budget (Wilcoxon signed-rank $p < 10^{-80}$, one-sided, across 531 basins).

The PGD-to-FGSM ratio grows monotonically with ε: 1.01× at $\varepsilon = 0.01$, 1.09× at $\varepsilon = 0.05$, 1.27× at $\varepsilon = 0.1$, and 1.71× at $\varepsilon = 0.2$ (Figure 1). At small budgets the model's response is approximately linear within the ε neighborhood, so a single gradient step nearly saturates the achievable damage; at larger budgets the response curves away from this approximation, and iterative refinement uncovers worse forcing-error pat-

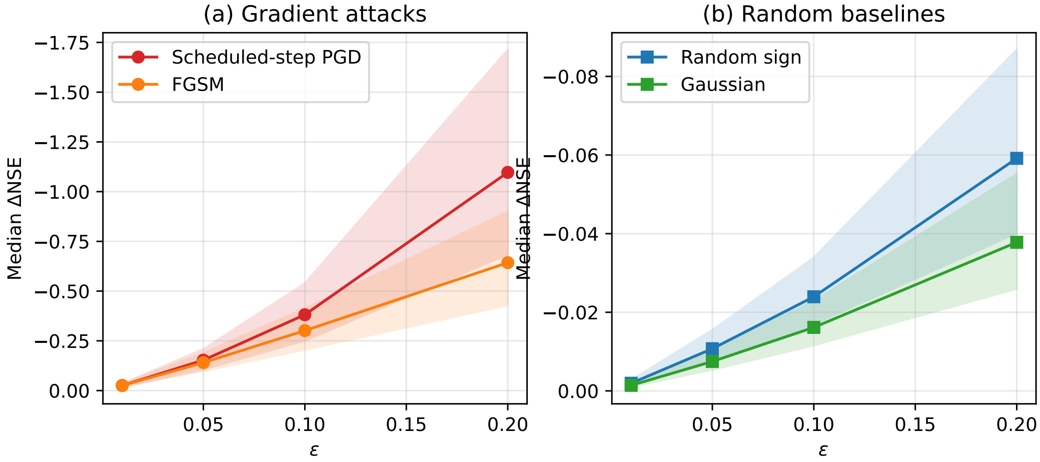

Figure 1 — Median ΔNSE as a function of ε for the gradient attacks and the random baselines, with IQR shown as shaded bands. (a) Adversarial attacks: scheduled-step PGD and FGSM. (b) Random-noise baselines. Note the different y-axis scales between panels.

terns than the single-step probe. The PGD-to-random-sign ratio also grows with ε (13.2×, 14.2×, 15.9×, 18.5×), so random-noise sensitivity analysis underestimates worst-case vulnerability most severely at the budgets that matter most for stress testing.

3.1.1 Distribution-Level View at $\varepsilon = 0.1$

Figure 2 shows the per-basin distribution of ΔNSE under scheduled-step PGD at $\varepsilon = 0.1$. The distribution is right-skewed: the 10th percentile is −0.755, the median is −0.381, and the 90th percentile is −0.179. The distance between the 10th percentile and the median (0.37) is about twice the distance between the median and the 90th percentile (0.20), so the median statistic alone hides a tail of more severely vulnerable basins.

The size of this tail differs sharply across attack types. The fraction of basins with adversarial NSE below zero is 15.4% (82 of 531) under scheduled-step PGD, 9.4% (50) under FGSM, and 0.6% (3) under Gaussian noise. Under the stricter $\Delta\text{NSE} < -1$ threshold, scheduled-step PGD produces severe degradation in 25 of 531 basins (4.7%). Even single-step FGSM identifies an at-risk population an order of magnitude larger than random noise. The below-zero fraction is, however, strongly skill-dependent: it ranges from 78.6% among the 14 basins with clean NSE below 0.3 to 2.8% among the 143 basins with clean NSE above 0.8. The gradient-versus-random ratio, by contrast, is stable across skill levels (9.7× to 16.2× across clean-NSE bins, and 15.4× even among the most skillful basins), so the conclusion that gradient attacks far exceed random noise is not an artifact of a few low-skill catchments. We therefore report median ΔNSE, not the below-zero fraction, as the headline statistic.

3.2 Vulnerability Tracks Hydrological Memory

The per-basin damage is not hydrologically random. No catchment class escapes: the median ΔNSE under scheduled-step PGD at $\varepsilon = 0.1$ is −0.38 across all 531 basins, and even the most robust class—the 31 rain-driven, flashy catchments (flow-duration-curve slope above 2)—still loses a median of 0.27. But severity varies systematically with catchment character. Relating per-basin damage to CAMELS catchment attributes (Addor

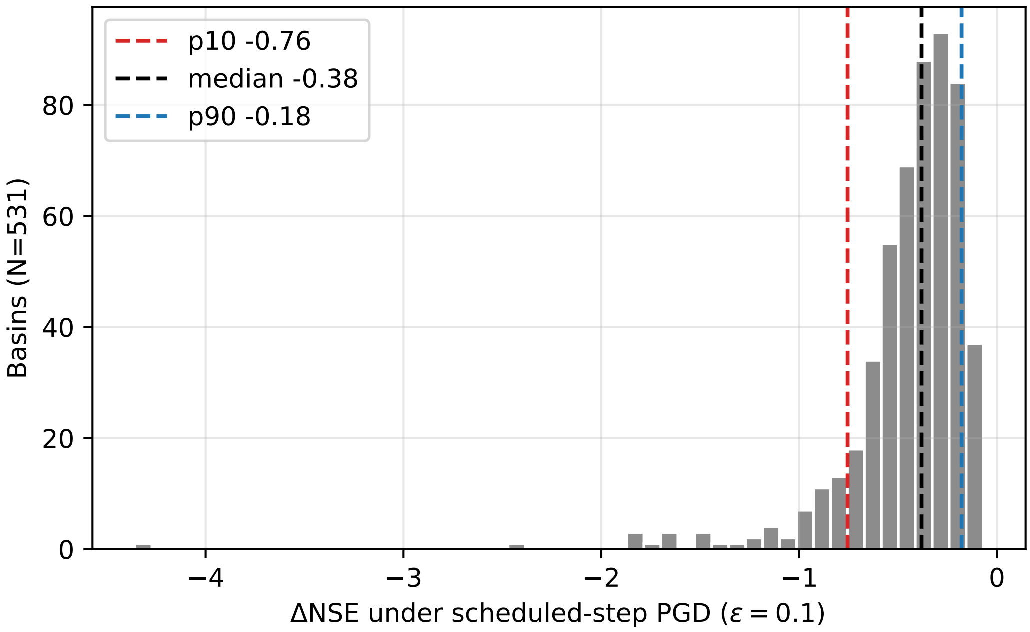

Figure 2 — Distribution of per-catchment ΔNSE under scheduled-step PGD at $\varepsilon = 0.1$ ($N = 531$ basins). Dashed lines mark the 10th percentile (−0.76), median (−0.38), and 90th percentile (−0.18).

et al., 2017), the single strongest predictor is the snow fraction of precipitation: more snow-influenced catchments are more vulnerable (Spearman $\rho = 0.52$), and the association *strengthens* after partialling out clean-data skill ($\rho = 0.61$ controlling for clean NSE, $p < 10^{-50}$, $N = 531$; Figure 3). Snow-dominated catchments (snow fraction above 0.2, $N = 152$) suffer a median ΔNSE of −0.51 against −0.32 elsewhere, despite being fit better on clean data (median clean NSE 0.77 versus 0.72). The vulnerability is thus not a restatement of low skill: the raw correlation between damage and clean NSE is only −0.26, and the snow signal survives controlling for it. A coherent gradient of catchment storage and memory accompanies it: damage rises with the mean half-flow date, mean elevation, and baseflow index, and falls with mean precipitation (all $|\rho| > 0.27$ after skill control). The most vulnerable catchments are therefore the snow-influenced, storage-dominated ones and the least vulnerable the flashy, rainfall-driven ones; we take up the mechanism this pattern suggests in the Discussion.

3.3 Which Forcing the Attack Exploits, and What It Does to the Hydrograph

If the vulnerability tracks hydrological memory, the attack should exploit the forcing that drives that memory. We test this on the 107-basin subset with two diagnostics of the untargeted attack at $\varepsilon = 0.1$: a per-forcing-variable attribution of the achieved perturbation, and its effect on the predicted flood hydrograph.

Attribution. We quantify how much of the adversarial damage each forcing channel carries by a leave-one-out measure: zeroing the achieved perturbation on one channel and re-evaluating the loss (temperature is treated as the single tied degree of freedom of the Maurer product). The channel attribution shifts systematically with snow regime (Figure 4a). In rain-dominated catchments precipitation dominates: it carries a

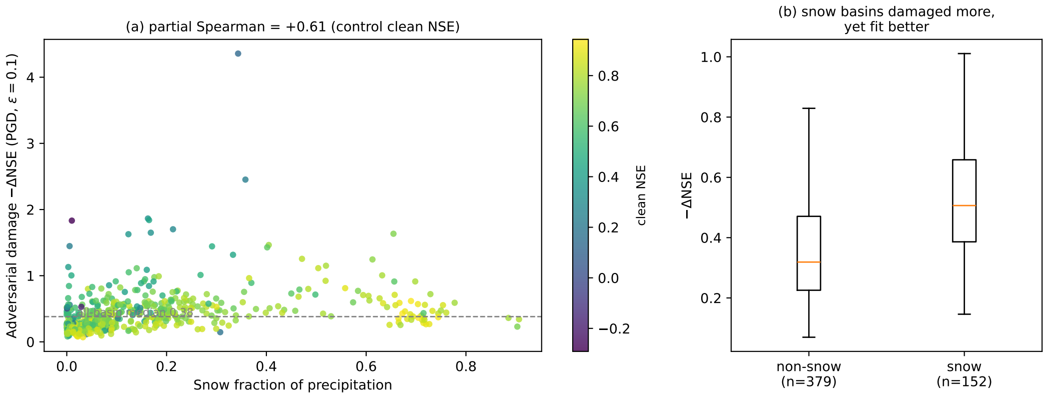

Figure 3 — Adversarial damage ($-\Delta$NSE, scheduled-step PGD, $\varepsilon = 0.1$) versus snow fraction across the 531 basins. (a) Per-basin damage rises with snow fraction; points colored by clean NSE, dashed line at the all-basin median. Partial Spearman $\rho = 0.61$ after controlling for clean NSE. (b) Snow-dominated catchments (snow fraction above 0.2) are damaged more than the rest, despite being fit better on clean data.

median 46% of the damage against 31% for temperature, and temperature is the leading channel in only 6 of 71 basins. In snow-influenced catchments the temperature channel gains leverage—its absolute leave-one-out attribution more than doubles (median 0.27 versus 0.11 for rain basins), while precipitation's absolute attribution is unchanged across regimes (0.21 versus 0.21). Temperature thereby rises to co-dominance in snow catchments (median share 0.45 versus 0.35 for precipitation) and is the leading channel in 23 of 36 of them. The temperature share rises with snow fraction at a partial Spearman correlation of +0.68 (controlling for clean NSE, $p < 10^{-15}$, $N = 107$). Because it is temperature *gaining* absolute leverage rather than precipitation losing it, the shift is not an artifact of the share normalization. Snow fraction is collinear with elevation and slope in CAMELS ($\rho = 0.70$ and 0.60), so the association attenuates but survives when elevation is held fixed (+0.39, $p < 10^{-4}$); we therefore read it as a snow-regime—physically, a snowmelt-temperature—effect that cannot be cleanly separated from the cold, high, steep character of these catchments.

Hydrological consequence. We next compare the clean and adversarial predictions around each observed flood peak (Figure 4b). The untargeted attack suppresses the predicted peak magnitude for the median basin by 11%, and about 2.4× more strongly in snow-influenced catchments (−18%) than in rain-dominated ones (−7%); this snow enhancement survives controlling for overall NSE damage (partial Spearman −0.24, $p = 0.01$). Suppression is the majority but not the universal response: 74 of 107 basins have their peaks damped (86% of snow basins against 61% of rain basins), while 33 have them amplified. The attack largely rescales peaks rather than relocating them—for 79% of peaks the predicted timing is unchanged and the local clean-versus-adversarial hydrograph-shape correlation exceeds 0.97—but the exception is telling: the 9 basins whose predicted peak timing does shift, by a median of 1–5 days, are *all* snow-influenced, the catchments where the attack can move the melt-driven peak as well as damp it. A minor water-balance signature accompanies this: the attack lifts the median water-year runoff coefficient from 0.37 to 0.44 and pushes a small fraction of snow-catchment water-years past the physical ceiling of unity (5% under attack versus 2% clean). This last effect is confined to catchments whose water balance already sits near that ceiling and is best read as tipping borderline balances over the line rather than manufacturing large volumes of water; it is reported with that caveat.

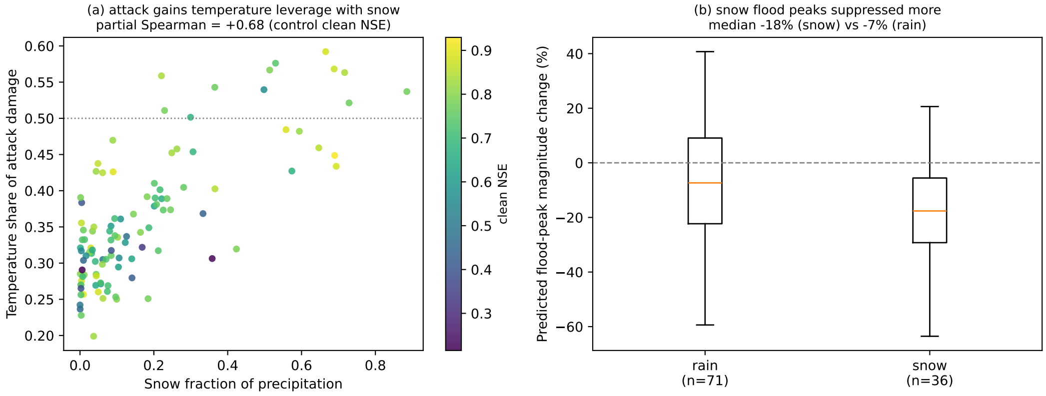

Figure 4 — Process attribution and hydrological consequence of the untargeted attack at $\varepsilon = 0.1$ (107-basin subset). (a) Share of adversarial damage carried by the temperature channel (leave-one-out attribution) versus snow fraction; points colored by clean NSE. Temperature gains leverage with snow (partial Spearman +0.68, controlling for clean NSE). (b) Change in predicted flood-peak magnitude, rain versus snow catchments; snow flood peaks are suppressed about 2.4× more strongly (median −18% versus −7%).

3.4 Statistically Constrained Perturbations Remain Effective

Table 4 presents the constraint-ablation results for scheduled-step PGD under the three plausibility levels. The physical constraint (non-negative precipitation, bounded temperatures and radiation) reduces median degradation by only about 3% relative to the magnitude-only baseline at $\varepsilon = 0.1$ (from −0.383 to −0.372), indicating that the most damaging adversarial forcing patterns are often already physically plausible: forcing the model into worst-case behavior does not require negative precipitation or unrealistic temperatures.

The statistical constraint, which additionally requires the perturbed forcing to preserve the per-variable whole-period mean and standard deviation of the clean series, reduces median degradation by 22% (to −0.299 at $\varepsilon = 0.1$), so it still retains 78% of the unconstrained damage. The reduction is modest because the constraint pins only the whole-period marginal moments: within the 270-day windows that the recurrent model actually consumes, the perturbation can still redistribute local mean and variance, and it remains free to alter temporal ordering and cross-variable structure.

The residual degradation under the statistical constraint is operationally significant and grows with ε: from −0.128 at $\varepsilon = 0.05$ to −0.299 at $\varepsilon = 0.1$ to −0.736 at $\varepsilon = 0.2$. The ratio of statistical-constraint to magnitude-only degradation stays near 0.7–0.8 across budgets. The practical implication is direct: forcing data that passes a routine QC checking only the climatological mean and standard deviation of each variable can still contain structured errors that substantially degrade model performance, and moment-based QC alone cannot rule out the existence of harmful forcing-error patterns within an observation-error-scale budget.

Table 4 — Constraint ablation results for scheduled-step PGD (untargeted) over the 107-basin subset. Values are median (mean) ΔNSE.

Constraint	N	ε=0.05	ε=0.1	ε=0.2
Magnitude	107	−0.156 (−0.172)	−0.383 (−0.448)	−1.095 (−1.602)
Physical	107	−0.145 (−0.164)	−0.372 (−0.429)	−1.050 (−1.565)
Statistical	107	−0.128 (−0.139)	−0.299 (−0.333)	−0.736 (−0.887)

3.5 Targeted Attacks and Pre-Event Vulnerability

Table 5 presents the results of targeted attacks. Flood-targeted attacks produce degradation comparable to, though slightly less than, untargeted attacks, while low-flow-targeted attacks are much weaker. The flood/low-flow contrast is about 6.8:1 by median ΔNSE, indicating that vulnerability is concentrated in high-flow regimes; the same ordering holds in ΔKGE.

Table 5 — Targeted attack results (scheduled-step PGD, $\varepsilon = 0.1$, magnitude constraint) over the 107-basin subset. Median [Q25, Q75]. ΔKGE is the physical-space Kling–Gupta change for the same adversary.

Target	N	Median ΔNSE [IQR]	Median ΔKGE [IQR]
Untargeted	107	−0.383 [−0.551, −0.241]	−0.155 [−0.231, −0.091]
Flood	107	−0.322 [−0.478, −0.225]	−0.149 [−0.216, −0.094]
Low-flow	107	−0.048 [−0.079, −0.016]	−0.016 [−0.055, +0.018]

Figure 5 shows the causal-trigger results. Restricting perturbations to progressively longer pre-event windows yields a monotonic increase in damage. Measured on the flood-peak predictions alone, the median $\Delta\mathrm{NSE}_{\mathrm{peak}}$ grows from −0.077 for a 1-day pre-event window to −0.148 (3 days), −0.229 (7 days), and −0.331 (14 days): perturbing only the fourteen days before each flood peak degrades the peak-flow predictions by a third of an NSE unit. Measured on the overall test-period NSE, the same restricted attacks cause median $\Delta\mathrm{NSE}_{\mathrm{overall}}$ of −0.014, −0.029, −0.049, and −0.075, so pre-event windows alone recover roughly a fifth of the unrestricted overall degradation. Both views identify the days immediately before flood peaks, the operationally critical forecasting window, as disproportionately consequential.

3.6 Safety Margins and Detectability

Two complementary analyses characterize how silently the model can be degraded: the minimum perturbation required to break it, and the detectability of harmful perturbations under standard distribution tests on the forcings.

The minimum-perturbation analysis identifies, for each basin, the smallest L_2 perturbation that drives test-period NSE below zero (Figure 6). Of 76 basins evaluated, 62 (82%) could be broken within the search budget, with a median minimum L_2 of 5.27 standardized units [Q25 4.14, Q75 6.09]. Because this L_2 is summed over the whole perturbed forcing series (about 1.6×10^4 effective timestep–feature components), it corresponds to a small per-component perturbation, of order 0.04 standardized units root-mean-square;

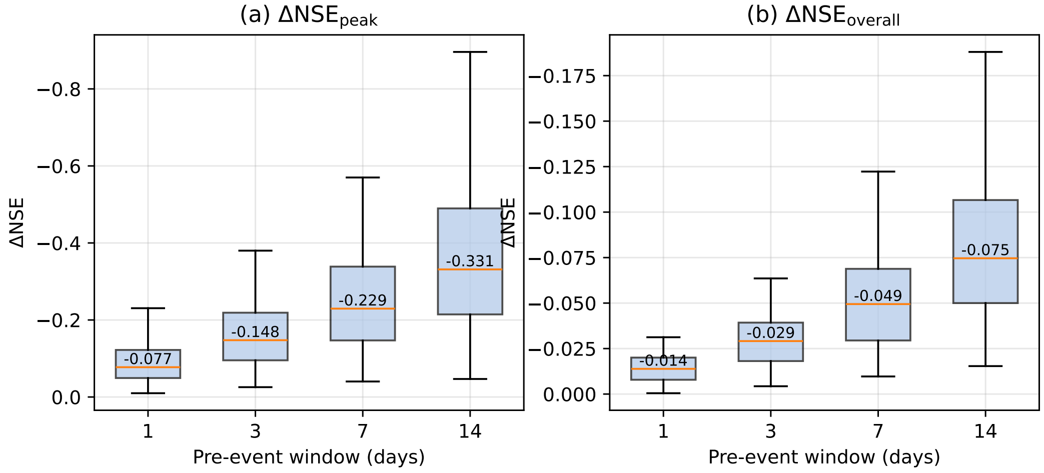

Figure 5 — Causal trigger: pre-event window length versus ΔNSE ($\varepsilon = 0.1$), measured on the flood-peak predictions ($\Delta\text{NSE}_{\text{peak}}$) and on the overall test-period NSE ($\Delta\text{NSE}_{\text{overall}}$). Box plots over the 107-basin subset; median values are annotated.

a coordinated perturbation far below the per-component ε budget is sufficient to break a majority of basins. The 14 basins not broken within the budget have wider safety margins, generally the more skillful catchments.

Detectability is assessed by a Kolmogorov–Smirnov (KS) two-sample test between the clean and perturbed whole-period marginal of each forcing variable for the magnitude-constrained adversary, paired with the achieved |ΔNSE| (Figure 7). Here the result is unambiguous in the opposite direction from a naive expectation: the unconstrained adversarial perturbation shifts the marginal of every forcing variable enough that the KS test rejects equality of distributions for all 107 basins (smallest-feature $p < 10^{-50}$ in every case), even though most of these same perturbations are physically plausible (Section 3.2). A whole-period distributional QC therefore *would* flag the unconstrained attack. The vulnerability that survives quality control is not the unconstrained attack but the moment-preserving one: by construction the statistical-constraint adversary leaves each feature's mean and standard deviation unchanged, so a moment-based QC cannot see it, yet it still retains 78% of the damage (Section 3.2). The operationally relevant gap is thus between moment-based QC, which the moment-preserving adversary evades by design, and full-distribution monitoring, which catches the unconstrained attack.

The two analyses together delineate where input-side defenses can and cannot help. An unconstrained adversarial perturbation is damaging but statistically conspicuous, so a distributional QC on the forcings is a meaningful defense against it. A moment-preserving perturbation is, by construction, invisible to the mean/standard-deviation checks that constitute routine QC, while retaining most of the damage; detecting it would require diagnostics that go beyond the feature-marginal moments, such as temporal autocorrelation checks, cross-feature consistency tests, or residual checks on the predicted streamflow against an independent reference.

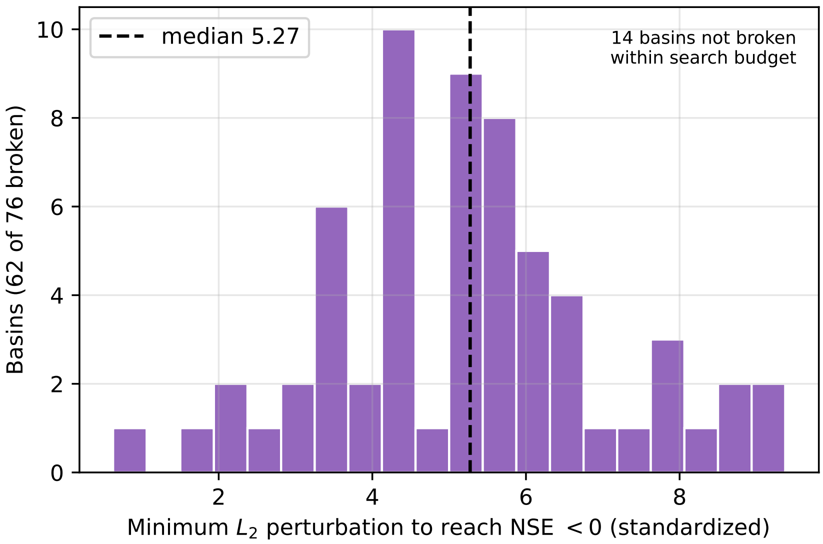

Figure 6 — Distribution of the minimum L_2 perturbation required to reduce NSE below zero, over the 76-basin subset. 62 of 76 basins are breakable within the search budget; the 14 unbroken basins are annotated.

4 Discussion

4.1 Relation to Prior Adversarial Assessment of Hydrological LSTMs

The closest precedent for this study is the FGSM-based robustness analysis of Yang et al. (2026), who applied single-step adversarial perturbations to LSTM and HBV models across 1,347 CAMELS-DE catchments and reported that LSTMs were more robust than HBV, that catastrophic failure was rare, and that model responses scaled approximately linearly with perturbation magnitude. Our CAMELS-US results qualify rather than contradict that picture, and the comparison is necessarily qualitative: the two studies differ in dataset (US vs. DE), attack (iterative scheduled-step PGD vs. single-step FGSM), budget, and metric, so we draw only direction-of-effect comparisons and avoid cross-dataset fragility rankings.

Two of the three qualitative claims of Yang et al. (2026) are reproduced here. Random noise remains a much weaker probe than gradient-based attack, and the response to single-step FGSM is approximately monotonic in ε across our budget range. The claim most sensitive to method choice is the third. On our victim, the iterative attack roughly doubles the FGSM degradation at the largest budget (PGD-to-FGSM ratio rising to 1.71× at $\varepsilon = 0.2$), so the severity of degradation, though not its qualitative existence, depends on whether an iterative method is used. This is consistent with the broader adversarial machine-learning literature, in which single-step methods systematically overestimate robustness (Carlini et al., 2019); our results transpose that pattern from image classification to a regression task on physical time series. We also note that our attack is, if anything, a conservative iterative method (no momentum, single restart), so the gap we report is a lower bound on what a stronger iterative attack could reveal.

A complementary comparison uses the Kling–Gupta Efficiency that Yang et al. (2026) reported. On our victim, the untargeted iterative attack at $\varepsilon = 0.1$ produces a median ΔKGE of -0.155 from a clean median KGE of 0.773, a relative degradation of about 20%. This is comparable to, if anything larger than, the relative KGE degradation in Yang

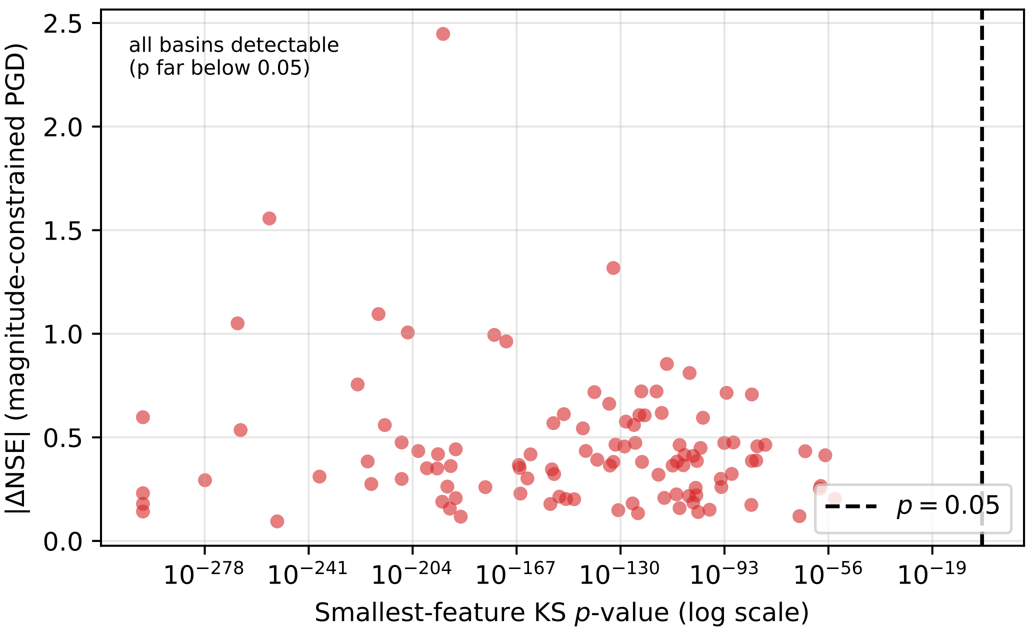

Figure 7 — Attack effectiveness (|ΔNSE|, magnitude-constrained scheduled-step PGD) versus statistical detectability (smallest-feature KS p-value) at $\varepsilon = 0.1$. Every basin falls far to the detectable side of $p = 0.05$ (red dashed line); the unconstrained attack is statistically conspicuous even where it is most damaging.

et al. (2026), despite our using a smaller normalized budget and an attack optimized for NSE rather than KGE. We treat this as a qualitative reference only, given the stacked differences in dataset, budget, attack, and target metric; it should not be read as a quantitative cross-dataset fragility ranking.

4.2 Mechanisms Underlying the Attack-Method Hierarchy

The hierarchy among the methods at $\varepsilon = 0.1$—scheduled-step PGD ahead of FGSM, with the random baselines an order of magnitude behind—admits a unified explanation. Random methods perturb each timestep and forcing variable independently and are exponentially unlikely to align with the directions of greatest model sensitivity; a best-of-K random search closes this gap only as $\sqrt{\ln K}$, so even extrapolating the random baseline to $K = 10^9$ samples leaves scheduled-step PGD about seven times more damaging (Supporting Information). FGSM uses the gradient at the clean operating point to identify an aligned direction but takes only one step, so where the model's response curves within the ε neighborhood, this direction is no longer the worst attainable. Scheduled-step PGD tracks the curvature iteratively.

The adversarial-to-random gap (15.9× at $\varepsilon = 0.1$) far exceeds the FGSM-to-PGD gap (1.27×), revealing the relative importance of the two methodological refinements: using gradient information at all matters far more than refining how that gradient is followed. The corresponding recommendation is asymmetric: a single-step gradient-based attack should be a minimum requirement for any future hydrological robustness assessment, whereas iterative refinement is the natural gold standard for deployment-grade evaluations.

The constraint-ablation experiment provides a further mechanistic clue, but it must be read carefully. Pinning each feature's whole-period mean and standard deviation removes only about a fifth of the damage, so most of the harmful signal does not live in the whole-period marginal moments. It is carried instead by structure those moments do not constrain: within-window redistribution of local mean and variance, temporal ordering of wet and dry days, and cross-variable correlations, all of which the recurrent state is sensitive to. We emphasize that whole-period moment preservation is a weaker constraint than per-window moment preservation; a constraint that pinned the moments of every input window would remove more damage, and our result bounds only what the whole-period version leaves available.

4.3 What the Vulnerability Reveals About the Learned Hydrology

The catchment-attribute analysis (Section 3.2) turns the attack into a probe of the model's learned dependencies. That adversarial damage scales with hydrological memory—snow fraction, half-flow date, elevation, and baseflow index—while the gradient-versus-random hierarchy itself is nearly independent of catchment type points to a single mechanism. Where streamflow skill rests on integrating forcing over long timescales, as in snowpack accumulation and storage-fed baseflow, the mapping from forcing to discharge is smooth and spread over many timesteps, giving a coordinated adversary a long lever: small, individually innocuous shifts distributed across the accumulation season compound into a large error in the integrated state. Where discharge is a fast, near-instantaneous response to recent rainfall, as in flashy rain-driven catchments, the same budget buys less, because fewer timesteps carry decisive information. That the model is *most* fragile in the snow catchments it fits *better* than average cautions against reading clean-data skill as robustness: high snow-catchment skill reflects a learned reliance on long-memory integration, which is precisely what the adversary exploits. The attribution and hydrograph analyses (Section 3.3) bear this out directly: in snow catchments the attack shifts its leverage onto the temperature forcing that drives snowmelt, and it preferentially suppresses—and, in the snowiest basins, retimes—the melt-driven flood peaks, the very pathway whose long memory the model has learned to integrate.

4.4 Implications for Operational Hydrological Deployment

These findings should not be read as evidence against operational deployment of LSTM rainfall-runoff models. The adversarial perturbations are diagnostic rather than predictive: they test whether harmful error structures exist within observation-error-scale uncertainty budgets, not whether those exact perturbations are likely to occur. A model that fails under such constrained stress tests cannot, however, be assumed robust on the basis of average-case random-noise experiments alone, because the stress test reveals harm that the random analysis is structurally unable to find.

A first implication concerns reporting conventions. We suggest that an adversarial ΔNSE at $\varepsilon = 0.1$ accompany clean-data skill summaries whenever a model is proposed for operational use. The marginal cost is modest—a single iterative pass per catchment over the test period—and the diagnostic power is qualitatively beyond what random-noise sensitivity analysis provides.

A second implication concerns input quality control. Routine QC built around marginal-moment checks is well-suited to detecting gross sensor failure, and our detectability analysis shows it would also flag an unconstrained adversarial perturbation, which shifts the forcing marginals conspicuously. Its blind spot is the moment-preserving perturbation, which by construction leaves the checked statistics unchanged while retaining most of the damage. Closing that blind spot requires higher-order input validation—temporal autocorrelation checks, cross-feature consistency tests, or model-side reconstruction residuals—rather than feature-marginal checks alone.

A third implication concerns the temporal structure of forecasting risk. The causal-trigger experiment shows that perturbations restricted to the fourteen days before a flood peak degrade the peak-flow predictions by a median of 0.33 NSE units and recover about a fifth of the unrestricted overall degradation. In operational flood forecasting this is precisely the window in which input data are most consequential and most often degraded by sensor failure or telemetry latency. Robustness budgets and data-quality monitoring should therefore be tightened around such pre-event windows rather than applied uniformly across the record.

4.5 Caveats and Avenues for Future Work

Several caveats bound the generality of these conclusions. The victim model is a single-layer LSTM with 256 hidden units and a fixed set of 27 static catchment attributes. Entity-aware LSTMs, sequence Transformers, and state-space architectures such as Mamba may interact with adversarial perturbations differently; a systematic architectural sweep on the same 531-basin benchmark is the natural next step. The dataset is CAMELS-US at daily resolution with the Maurer forcing, in which the minimum and maximum temperature inputs are identical; sub-daily datasets, other forcings, and non-CONUS hydroclimates may shift both the median and the tail of the vulnerability distribution. Our iterative attack is deliberately conservative (no momentum, single restart), so the degradations we report are lower bounds on what a stronger attack could achieve.

Defense strategies were intentionally outside the scope of this study. Adversarial training, ensemble averaging across model seeds, and input-side denoising each make a different methodological commitment, and a rigorous evaluation of these strategies—ideally against the same multi-level stress test rather than against FGSM alone—is a natural follow-up. The fact that statistically constrained perturbations retain most of the unconstrained damage suggests that input-side denoising is unlikely to suffice on its own.

A final interpretive caveat is the diagnostic-not-predictive framing noted above. The adversarial perturbations provide a worst-case bound within an observation-error-scale budget; they do not estimate the empirical frequency of such forcing-error patterns. Their *magnitude*, however, is realistic. Comparing the $\varepsilon = 0.1$ budget to the day-to-day disagreement among the Daymet, Maurer, and NLDAS forcing products over the same 531 basins and test period, the precipitation perturbation (± 0.70 mm/day) is only 0.45 times the median inter-product spread (1.57 mm/day), and the temperature perturbation ($\pm 1.07\,^{\circ}$C) is comparable to it (1.36 times the 0.79 °C spread); the three products' daily range exceeds the budget on 52% of days for precipitation and 70% for temperature. Errors of this size are therefore routine in real gridded forcing. What is not realistic is their *structure*: the damage comes from the adversarial temporal coordination of these individually plausible errors, which ordinary product disagreement does not supply. Closing the remaining gap would require coupling the attack to a calibrated input-error model or comparing the adversarial ΔNSE distribution to that induced by historical input-error events; both are open directions.

5 Conclusions

We investigated how the choice of perturbation method affects adversarial robustness conclusions for a standard LSTM rainfall-runoff configuration, evaluating methods spanning the spectrum from random noise to iterative gradient-based optimization across the 531-basin CAMELS-US benchmark. The experiments support five principal findings. First, attack-method choice is decisive: iterative adversarial perturbations produce a median degradation about sixteen times that of structure-matched random noise of equal sampling budget at $\varepsilon = 0.1$, and even single-step FGSM, while weaker than the iterative attack at larger budgets, far exceeds any random baseline. Second, the gradient-versus-random gap is not a low-skill artifact: it is stable across clean-skill levels and remains about fifteenfold (15.4×) even among the most skillful catchments, while the below-zero failure fraction (15.4% of basins at $\varepsilon = 0.1$) is strongly skill-dependent and is reported only as a secondary statistic. Third, perturbations that preserve the per-variable mean and standard deviation of the forcing retain 78% of the unconstrained damage while evading moment-based quality control by construction, indicating that routine moment-based QC is insufficient to rule out harmful forcing patterns. Fourth, pre-event data quality is disproportionately important: perturbations confined to the fourteen days preceding a flood peak degrade the peak-flow predictions by a median of 0.33 NSE units, in the operational forecasting window where input data are most consequential and most often degraded. Fifth, this vulnerability is universal across catchments but its severity scales with hydrological memory: snow fraction is the strongest catchment predictor of adversarial damage (rank correlation 0.61 after controlling for clean-data skill), and the snow-dominated catchments the model fits better are damaged most, so clean-data skill alone is not a reliable indicator of adversarial robustness. In these catchments the attack gains leverage on the temperature forcing that drives snowmelt and preferentially suppresses their predicted flood peaks, exposing the learned snow-memory pathway as the least robust.

Taken together, these findings indicate that robustness assessment for standard hydrological deep-learning configurations requires multi-level stress testing that extends beyond random noise and single-step adversarial probes. We recommend iterative adversarial evaluation at observation-error-scale budgets as a standard diagnostic component of hydrological stress-testing workflows. Broader architectural generalization, evaluation of defense strategies, and the coupling of adversarial bounds to calibrated forcing-error models remain natural follow-ups.

Open Research

All code for adversarial attacks, evaluation, and analysis is available at `https://github.com/[repository]`. The LSTM model was trained using the open-source neuralhydrology framework (Kratzert et al., 2022). CAMELS-US data are publicly available from the USGS (Newman et al., 2015; Addor et al., 2017).

Acknowledgments

[Acknowledgments to be added.]

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